

1. Introduction

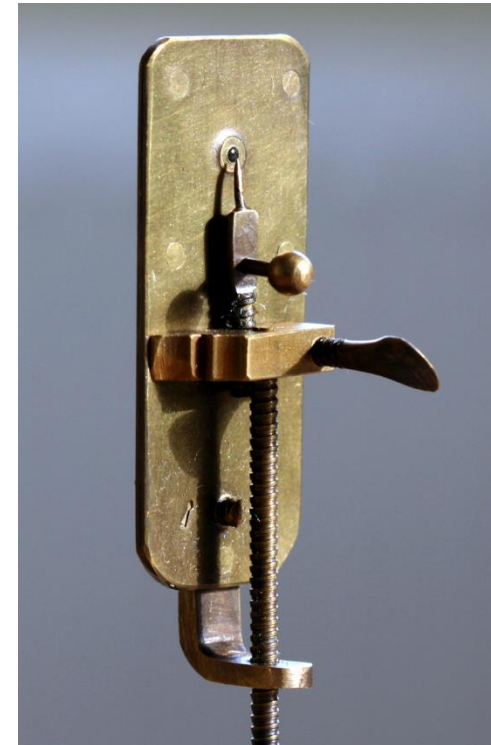
2. The Optical Microscope



Antonie van Leeuwenhoek
(列文虎克)

24 October 1632 – 26 August 1723

- Develop of the first conventional light microscope in the 15th century
- Magnification: about 200×
- Since then, a race to improve the magnification and quality of the images began, such as lens design, lens configuration, and new methods of sample preparation





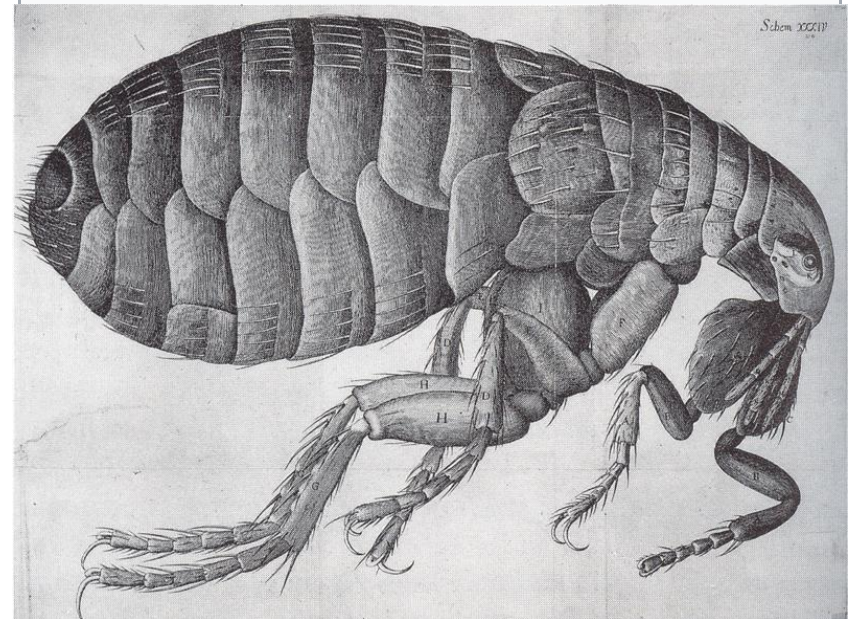
Robert Hooke (罗伯特·胡克)

1680 portrait conjecture
to be Hooke

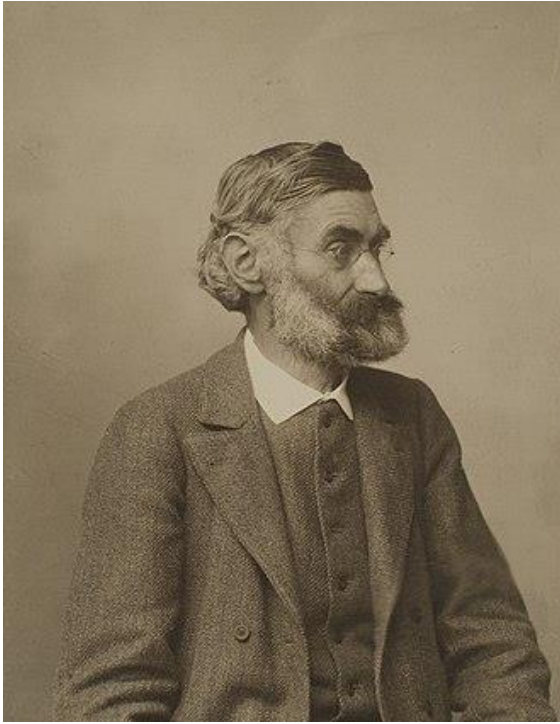
Born	28 July [O.S. 18 July] 1635 Freshwater, Isle of Wight, England
Died	March 3, 1703 (aged 67) London, England
Nationality	English
Alma mater	Wadham College, Oxford
Known for	Hooke's law Microscopy Coining the term 'cell'



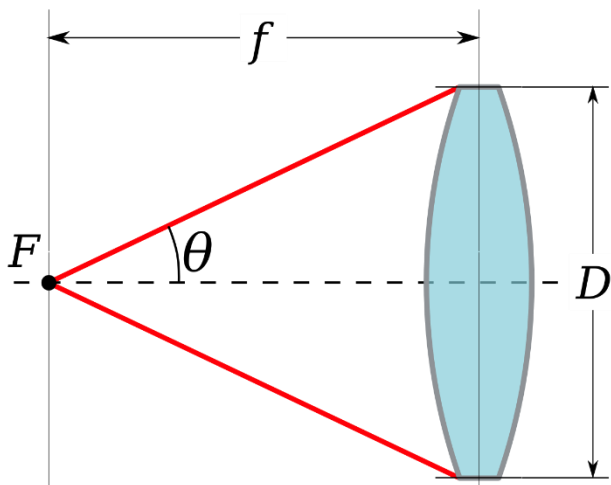
Robert Hooker' microscope



Hooke's drawing of a flea



Ernst Karl Abbe
(1840 – 1905)



German optician Ernst Abbe in 1873, found one of his most important conclusions: the relationship between the wavelength of light and the resolving power of a microscope, given by

$$d = \frac{0.612\lambda}{n \sin\theta}$$

d is the resolution, λ is the wavelength of the illuminant, n is the index of refraction of the medium between the specimen and the objective lens, and α is the half-angle (in radians) of the cone of light from the specimen plane accepted by the objective. The product $n \cdot \sin\theta$ is often expressed as the numerical aperture (NA) of the objective lens

A simple version:

$$d = \frac{\lambda}{2NA}$$

NA is called numerical aperture, 0.1 ~ 1

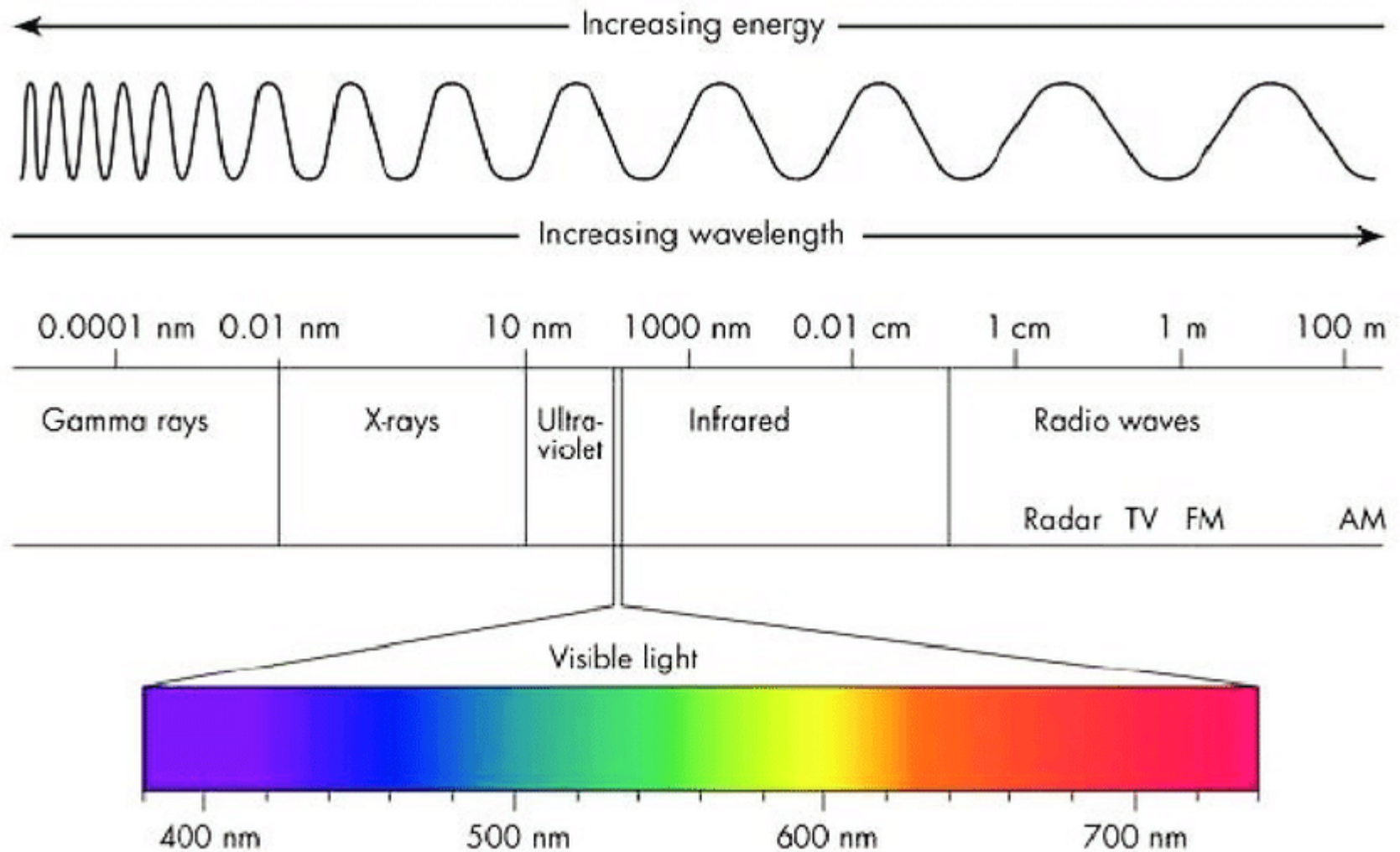
Homework 2

- How to obtain the resolution of an optical microscope as follows

$$d = \frac{0.612\lambda}{n \sin\theta}$$

•

$$d = \frac{\lambda}{2NA}$$



In later 19th century, J.J. Thomson demonstrated the origins of electron sources and also the possibility of forming an electron beam using electrical and magnetic fields.



J.J. Thomson

(1856 –1940)

Nobel Laureate in Physics, credited
with the discovery of the electron

In 1924, De Broglie established the relationship between a moving electron and the existence of a radiation with an associated wavelength, given by



Louis de Broglie (德布罗意)

(1892 – 1987)

Nobel Laureate in Physics, 1929

$$\lambda = \frac{h}{mv}$$

λ is the wavelength of the moving particle, h is Planck's constant, m is the mass of the electron, and v is the velocity of the electron.

- This equation implies that the wavelength depends only on the acceleration voltage of the electron.
- According to his experimental measurements, the effective wavelength of an electron beam is many orders of magnitude smaller than the wavelength of light.

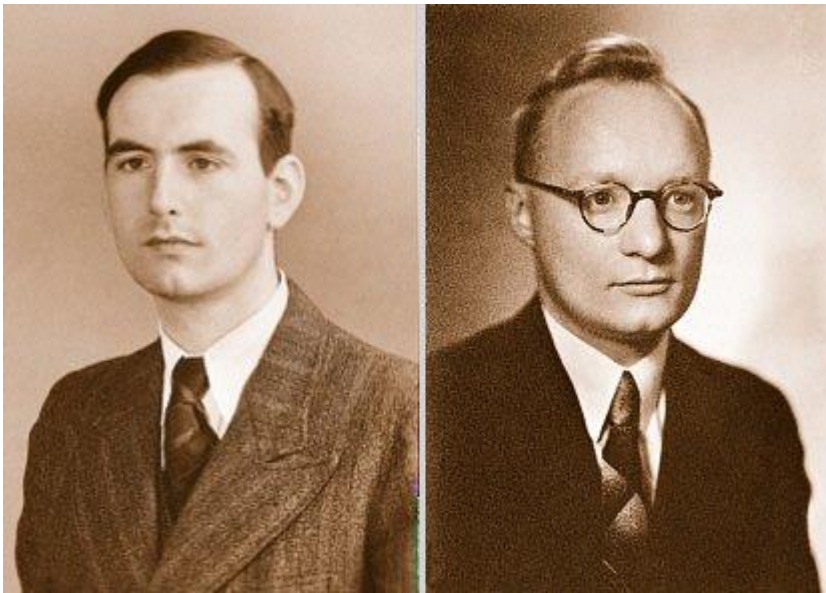
In 1926, Busch discovered the similitude (相似性) between the diffraction of electron beams through a magnetic field and the diffraction of light through a glass lens, concluding that geometric optic laws are also applicable to electron systems



Hans Busch
(1884–1973)

A German physicist.
He was a pioneer of electron optics and laid the theoretical basis for the electron microscope.

The first developed electron microscope was the transmission electron microscope (TEM) invented by Knoll and Ruska in 1932; a few years later, in 1936, the first commercial application of a TEM was released.



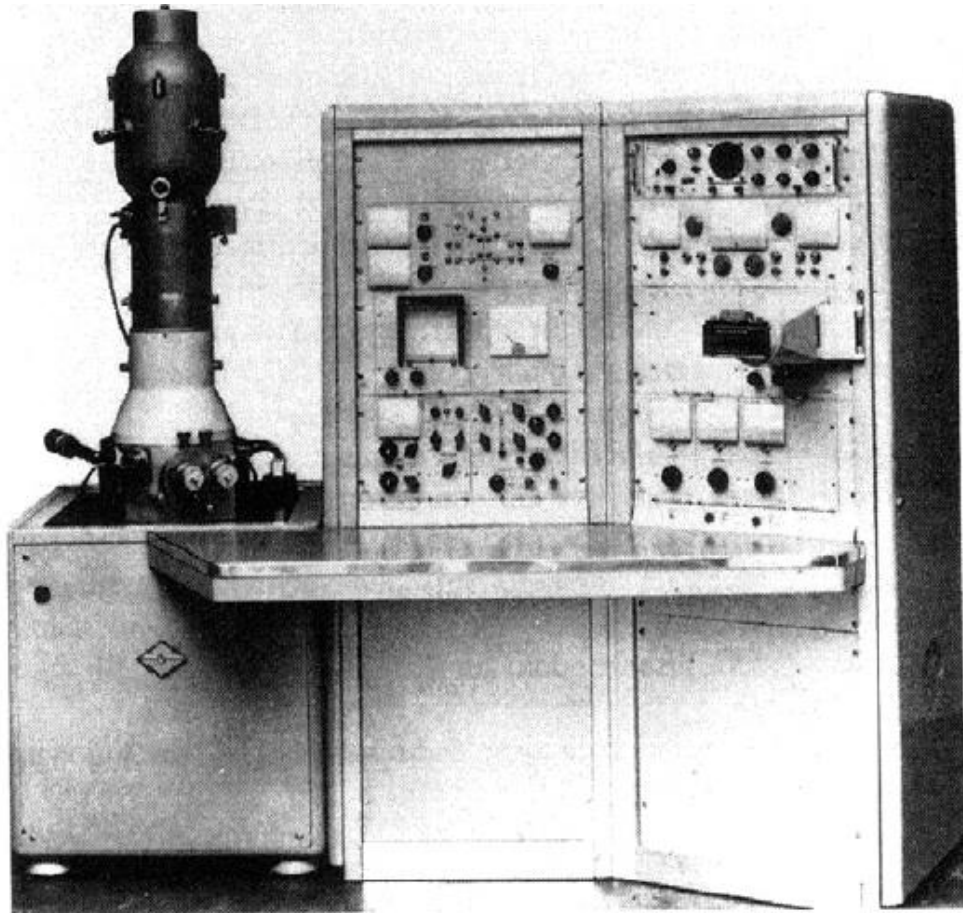
Ernst Ruska and Max Knoll

Ernst Ruska won Nobel Prize in Physics in 1986 for his work in electron optics, including the design of the first electron microscope



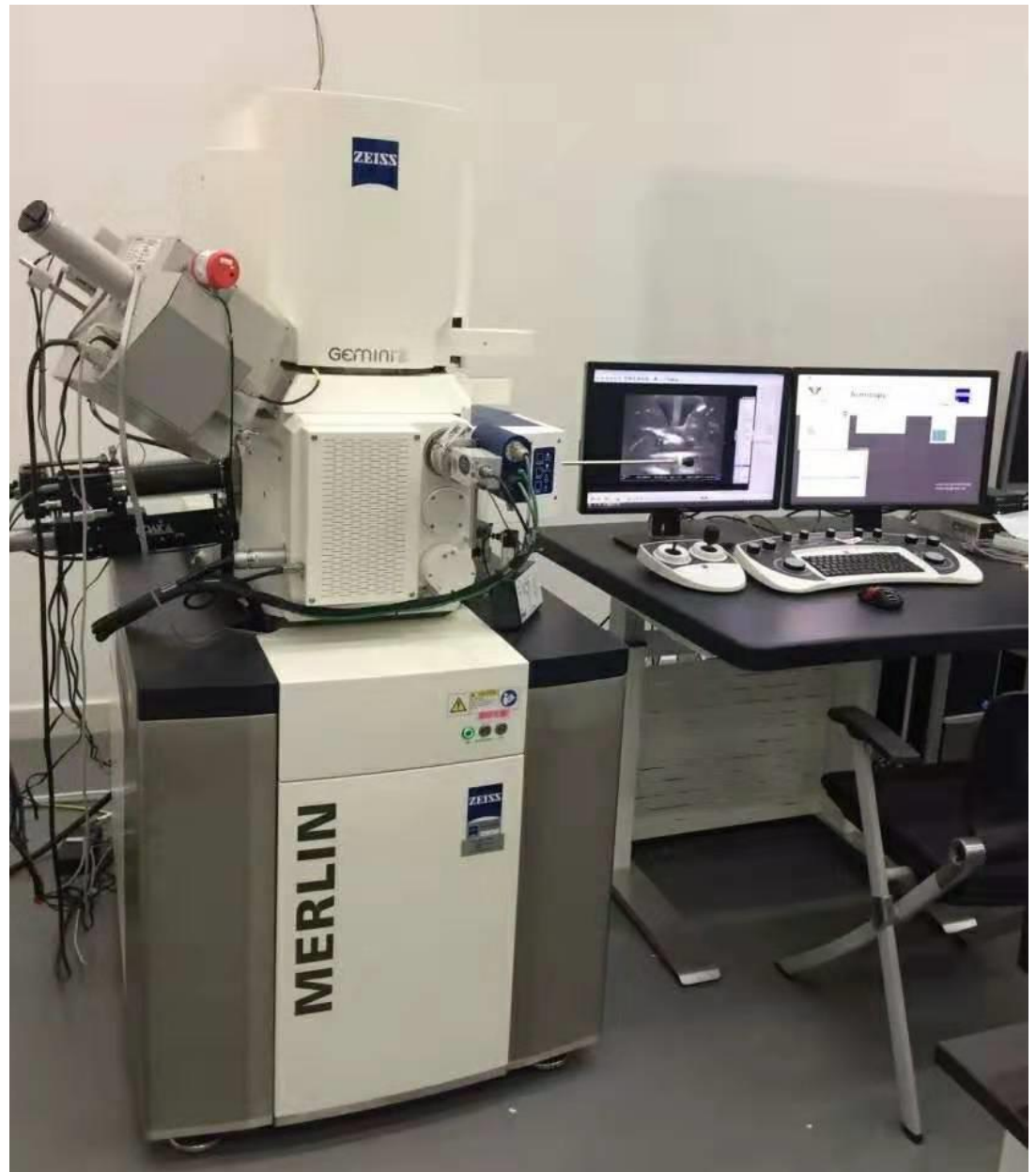
Electron microscope constructed by Ernst Ruska in 1933

- In 1965, the modern commercial **scanning electron microscope (SEM)** appeared from Cambridge Instruments
- its popularity increased with the years because of lower cost, simplicity of operation, and capabilities compared to TEM.
- Today, SEM is one of the most powerful characterization tools for diverse scientific fields and different research areas.



Cambridge Instrument Company: prototype of the first SEM

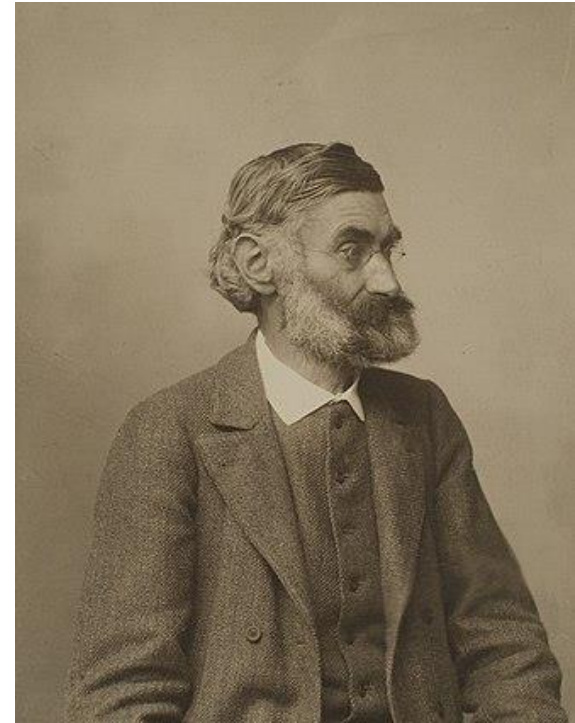
**SEM
ZEISS
Merlin**



SEM at SUSTech



Carl Zeiss
(1816 – 1888)



Ernst Karl Abbe
(1840 – 1905)

**TEM(equipped
with EELS)
Titan ETEM G2**



TEM at SUSTech

Resolution

- Generally speaking, TEM has better resolution than SEM due to the higher accelerate voltage (typically 100-300 kV)
- The best resolution for leading TEM companies:
FEI 63 pm (0.63Å)、 JEOL 58 pm、 Hitachi 78 pm

From
$$d = \frac{0.612\lambda}{n \sin\theta}$$

In vacuum, n is 1; $\theta_{\max}=90^\circ$; 
$$d_{lim} \approx \frac{1}{2} \lambda$$

From matter wave
$$\lambda = \frac{h}{mv}$$

v is the velocity of the electron

thus, λ is determined by the accelerated voltage

Resolution

Combined with theory of relativity

$$eU = mc^2 - m_0c^2 \quad U \text{ is the accelerated voltage}$$

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

Based on $\lambda = \frac{h}{mv},$


$$\lambda = \frac{h}{\sqrt{2m_0eU\left(1 + \frac{eU}{2m_0c^2}\right)}} = \frac{12.25}{\sqrt{U(1 + 10^{-6}U)}}$$

Resolution

$$\lambda = \frac{12.25}{\sqrt{U(1 + 10^{-6}U)}}$$

When U is

100 kV, λ is 0.0370 Å

200 kV, λ is 0.0251 Å

300 kV, λ is 0.0197 Å

1000 kV, λ is 0.0087 Å

U increased by 10 times, λ only increased by 4.25 times

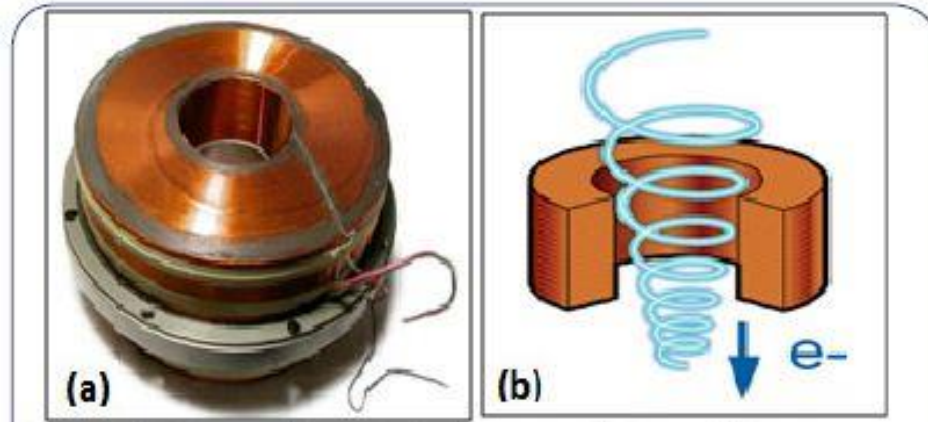
➤ The best resolution for leading TEM companies:

FEI: 0.63Å、 JEOL: 0.58Å 、 Hitachi 0.78Å

Resolution

- Obviously, the wavelength is not the main factor affecting the resolution of TEM.
- At present, we can't make electron microscopes with wavelength-limited resolution because we can't make perfect magnetic lenses like optical microscopes.

Magnetic lenses:



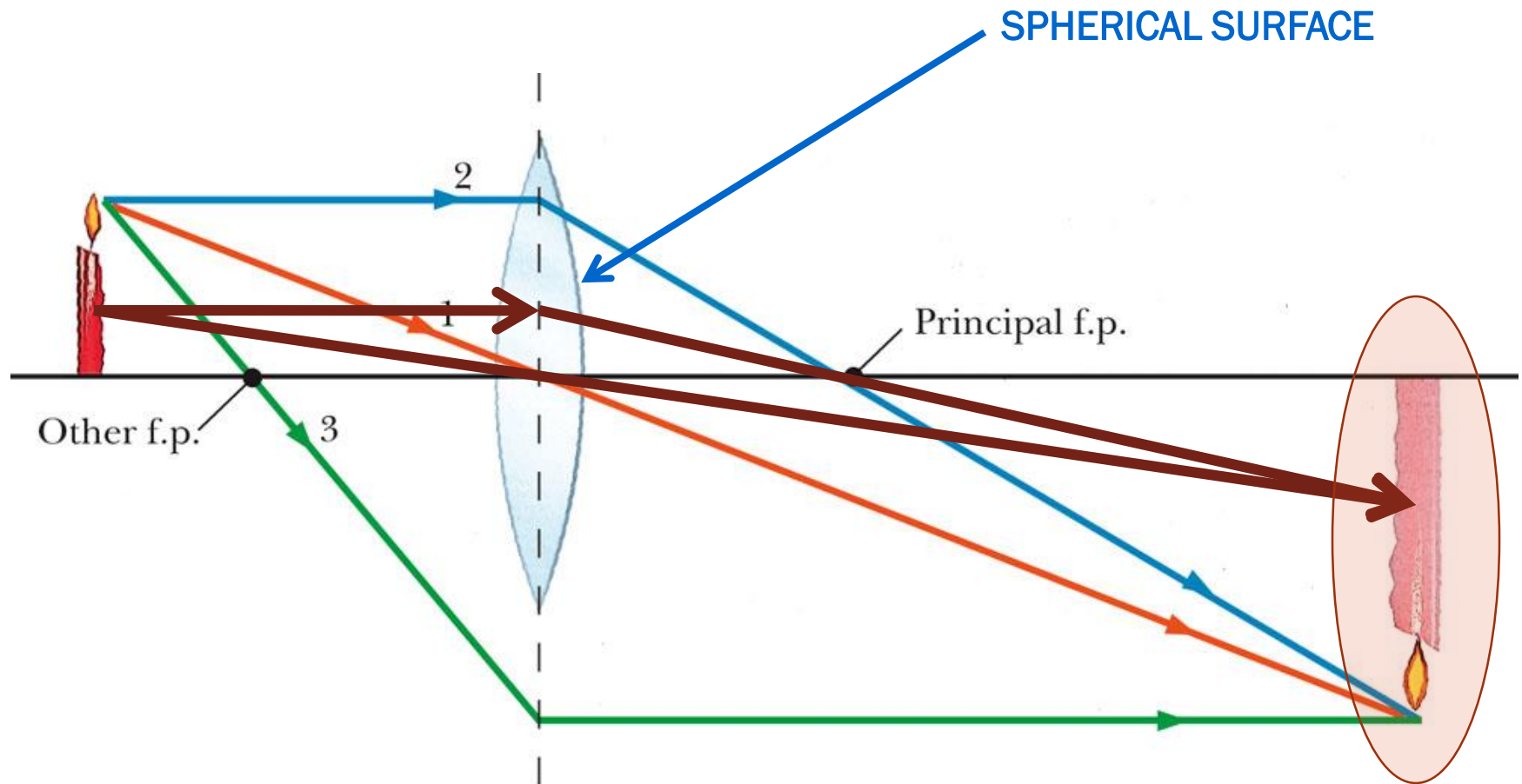
- This huge difference is caused by the aberrations of magnetic lenses. Therefore, several other electron microscopes with low acceleration voltage have higher resolution through aberration correction.
- Various aberrations (spherical aberration, chromatic aberration, astigmatism, distortion) of the magnetic lens, especially the influence of the spherical aberration on the numerical aperture of the objective lens, prevent the improvement of the resolution of the electron microscope.

Introductory Comments on the Optical Microscope



<http://www.microscopeworld.com/MSWorld/parts.aspx>

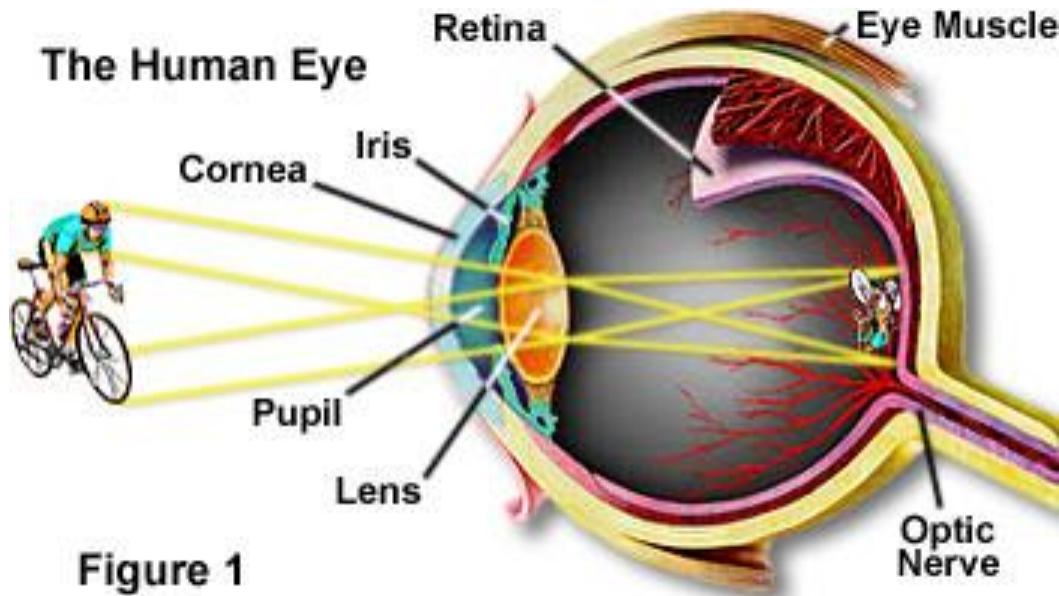
Simple Lens



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Optics Reviewed

- **How does the eye create an image for the brain to understand?**



The man on the bicycle appears upside down due to lenses, because lenses invert the image.

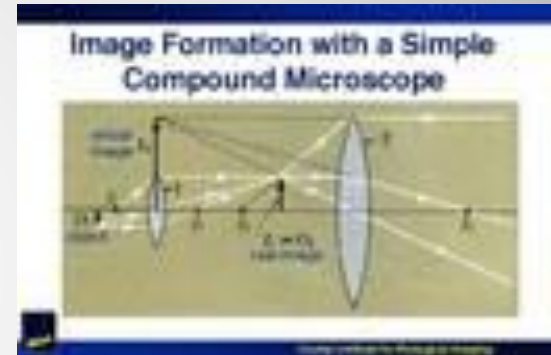
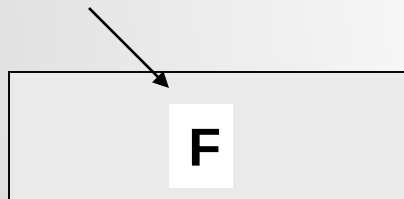
- If the ocular (eyepiece) lens has a magnification of 10x and the objective lens being used is 100x, the total magnification would be

1,000 x

Total Magnification Example:

- Images viewed through the eyepiece of compound microscopes will appear upside-down and backwards.

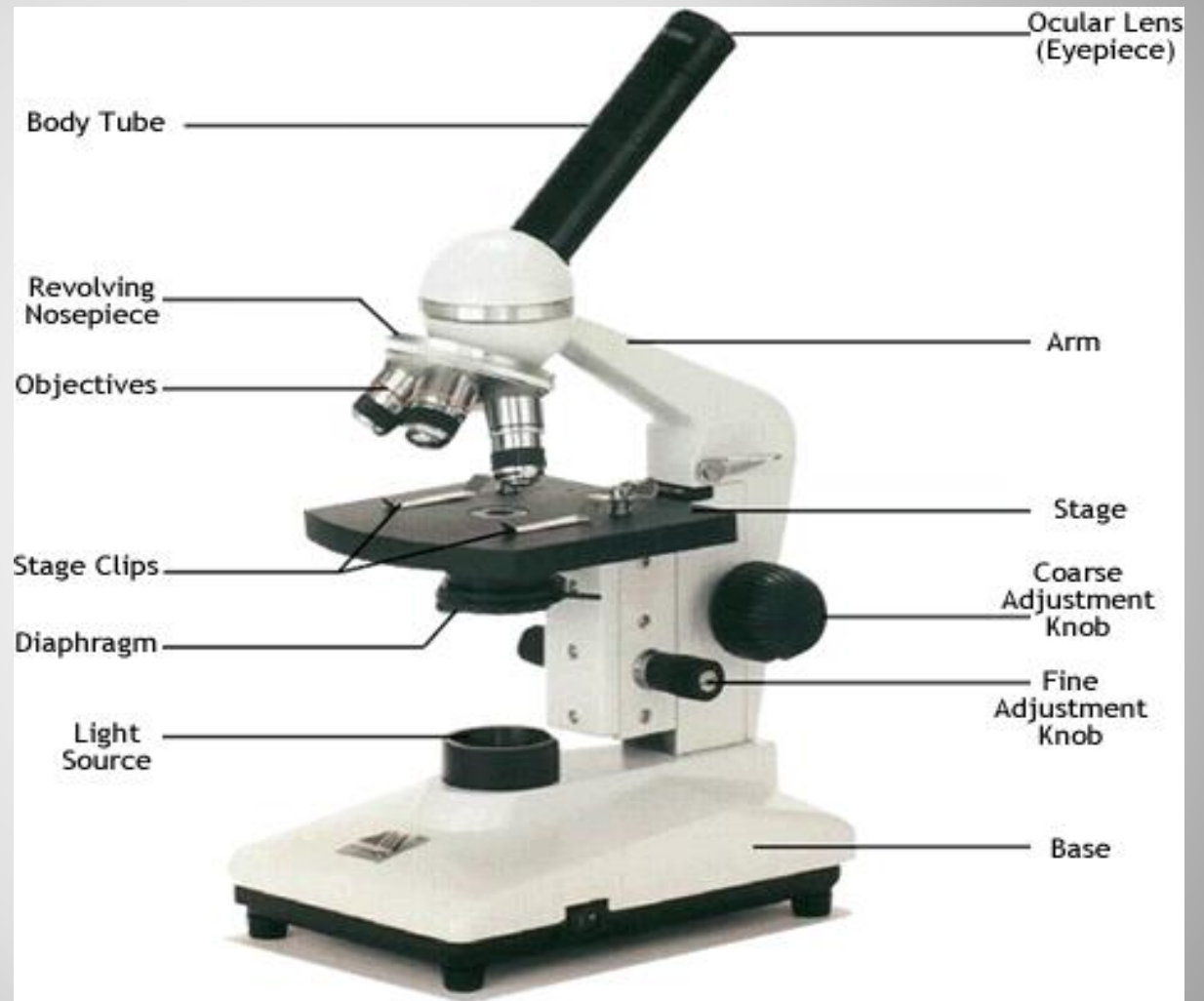
Slide with the letter “F”



The letter “F” as it appears when viewed through the eyepiece

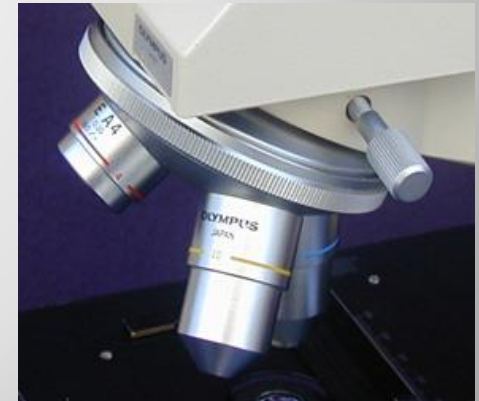
Optics (Continued)

Major Parts of the Optical Microscope

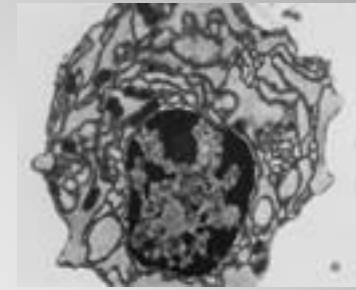


Important Microscope parts include :

- Ocular lens or eyepiece is used for viewing.
- Revolving nosepiece(旋转鼻架) contains objective lenses that are used to magnify the image in combination with the ocular lens.



Total Magnification:

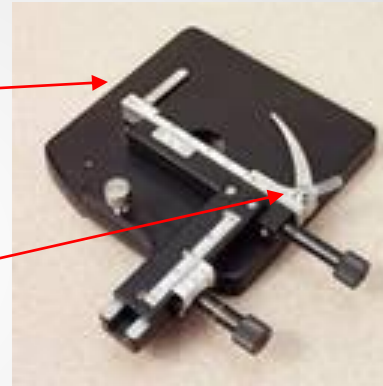


- The total magnification of the specimen being viewed is calculated using the ocular lens multiplied by the objective lens.
- For example, if the ocular lens is 10x and the ocular lens is 45x then the total magnification would be 450x.

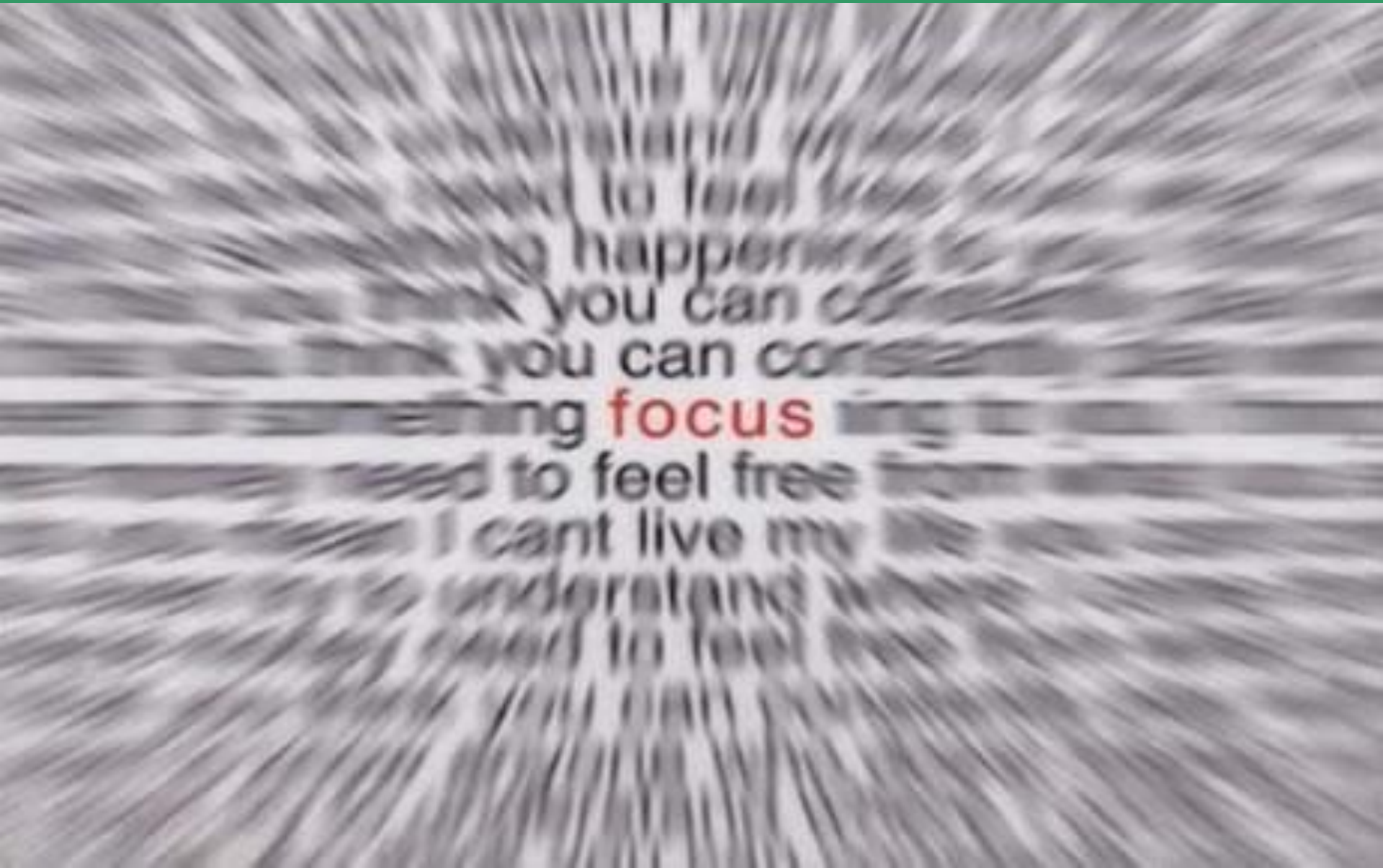


Other important microscope parts:

- **Stage-** the location of the specimen to be viewed
- **Clips-** utilized in holding the specimen in place
- **Lamp-** typically a light source underneath the stage
- **Diaphragm(光圈)**- controls the amount of light allowed to pass through the specimen



GETTING IN FOCUS



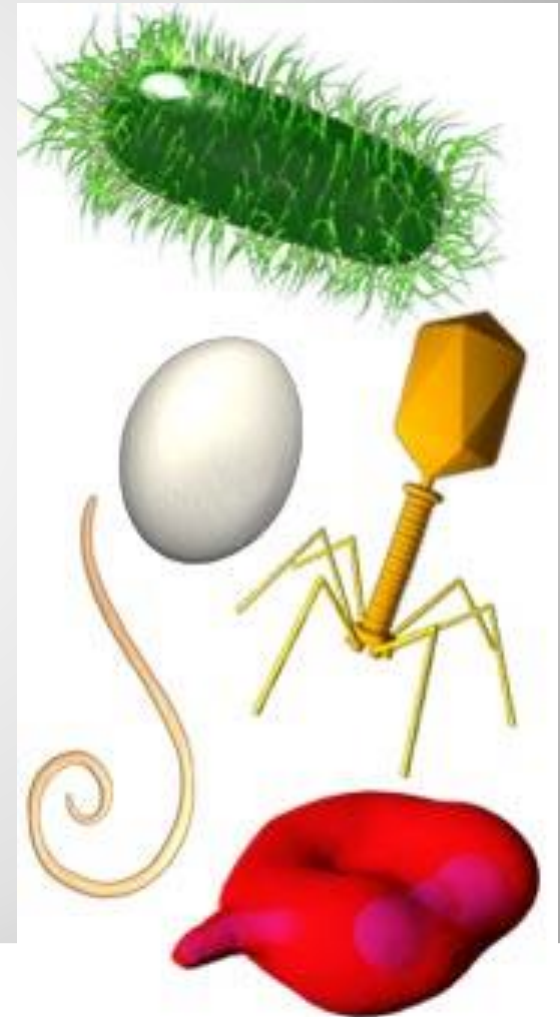
Focus and Resolution Parts

- **Coarse-adjustment knob-** is the larger of the two knobs. It is used in bringing the object into quick focus. →
- **Fine-adjustment knob-** is used for improving the clarity of the image, especially when viewing under high power. →



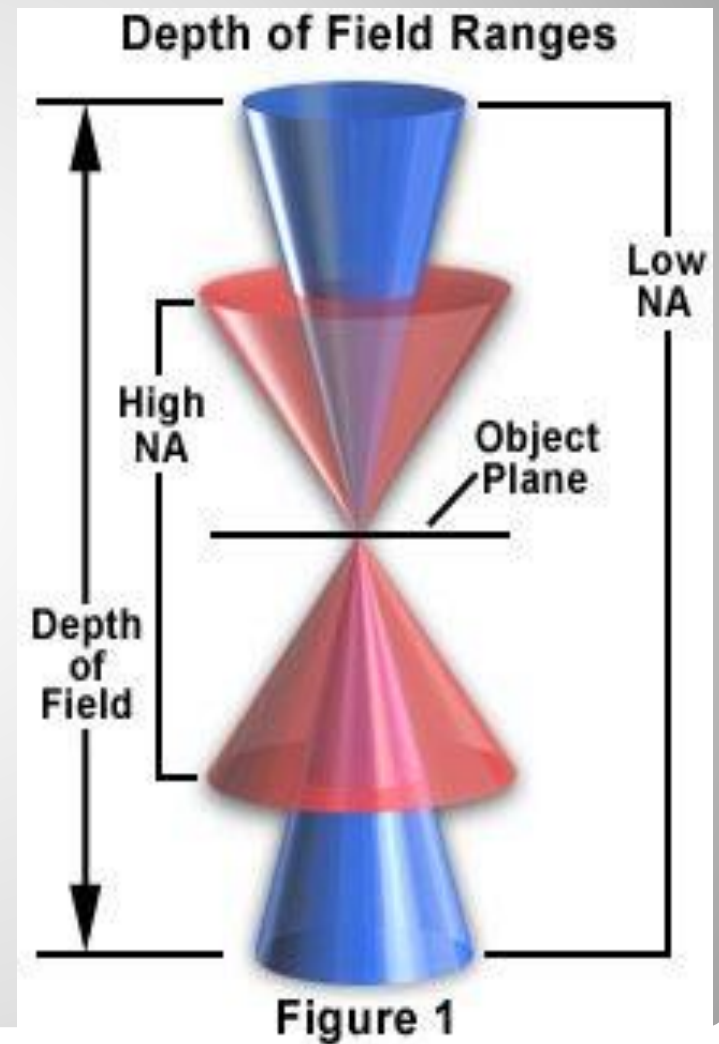
Factors that influence the quality and size of image:

- **Focal Length**
- **Size of sample**
- **Type of sample**
- **Quality of Microscope and lenses**
- **Amount of light on the sample**
- **Quality of sample**

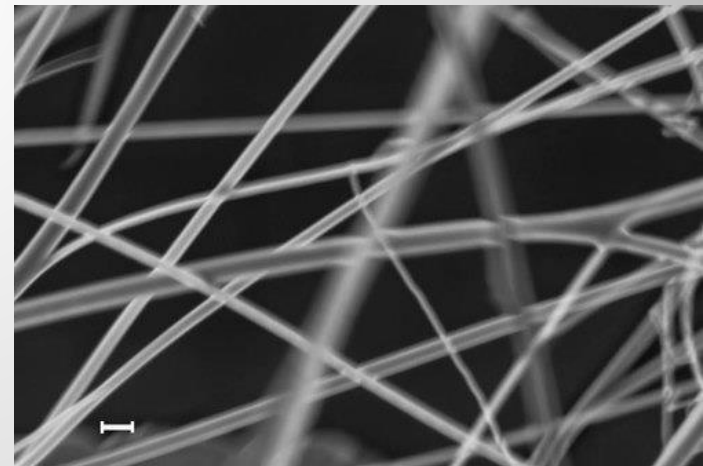
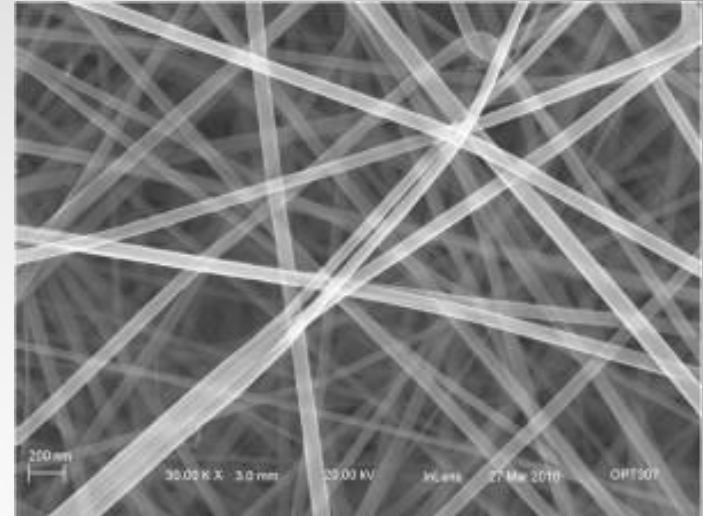


Depth of Field

- "Depth of field" refers to the thickness of the plane of focus.
- It is the vertical distance (from above to below the focal plane) that yields a useful image.



- With a large depth of field, all of the threads can be in focused at the same time.
- With a smaller or narrower depth of field, only one thread or a part of one thread can be focused, everything else will be out of focus.

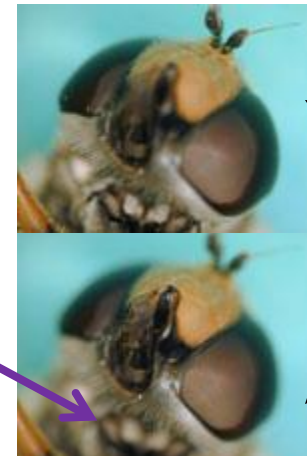
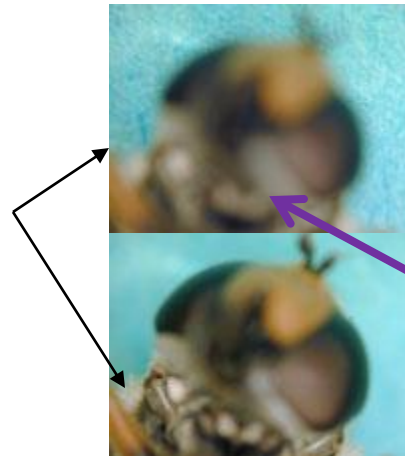


Depth of Field Example:



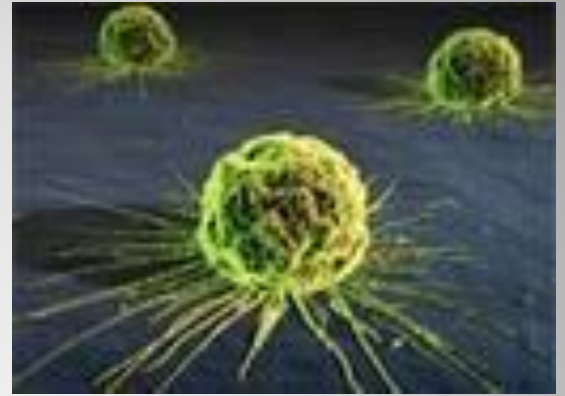
The series of images show how the depth of field can influence the appearance of an image.

With narrow depth of field, only part of the image is in focus at the same time.

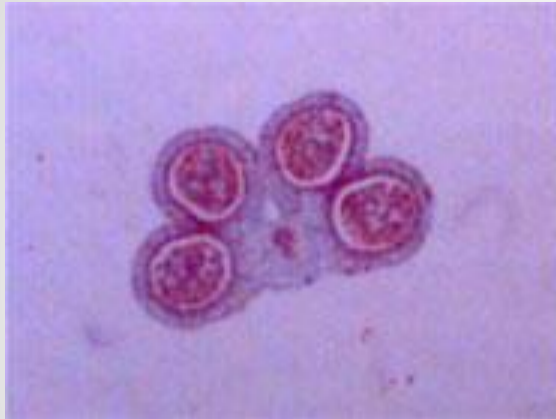


With a large depth of field, the entire image is in focus at the same time.

Resolution



- The amount of detail you can see in an image.
- You can enlarge a photograph indefinitely using more powerful lenses, but the image will blur (模糊) together and be unreadable.



Higher resolution



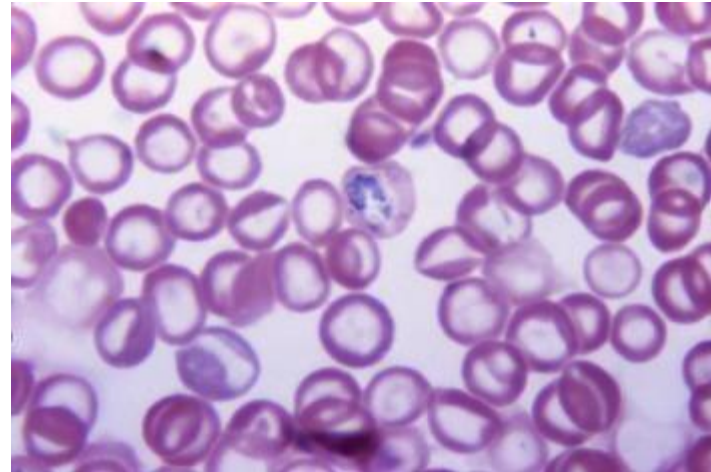
Lower resolution

Resolution

OBSERVATION MODES

(A) STAINING

Cell staining is a technique that can be used to better visualize cells and cell components under a microscope. By using different stains, one can preferentially stain certain cell components, such as a nucleus or a cell wall, or the entire cell.

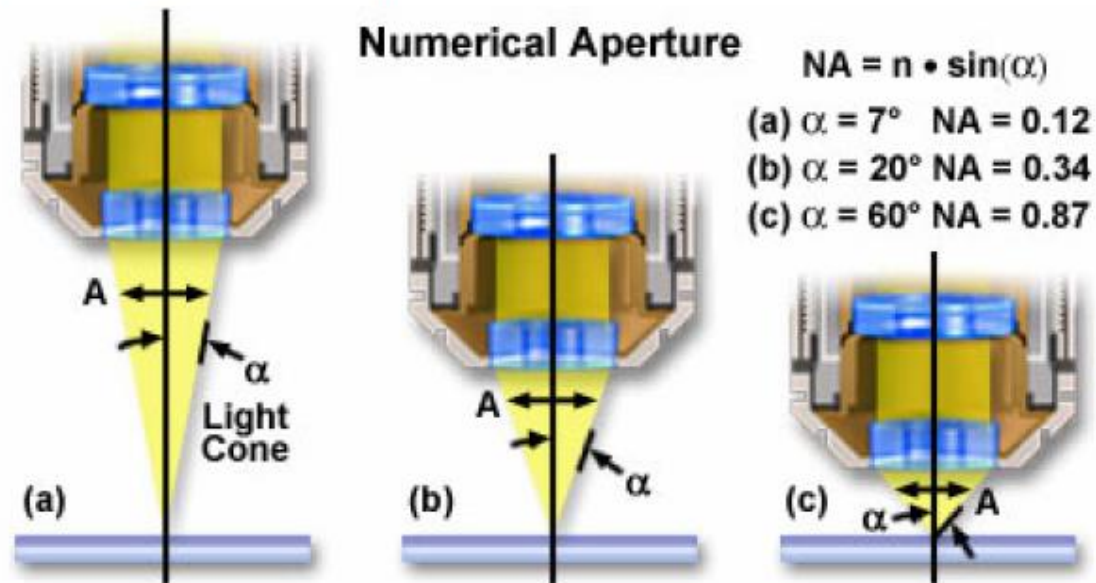


Light Microscopy

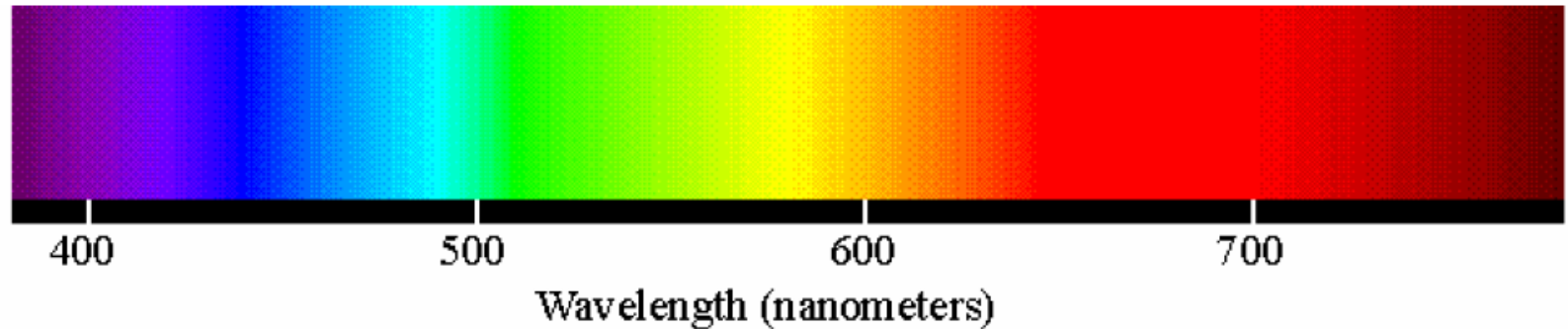
- Numerical Aperture, NA
 - Ability of the lens to collect light coming off of a specimen

$$NA = n \sin \theta$$

- n = index of refraction
- θ = half the angle of acceptance



Light Microscopy



- Resolution, R
 - Minimum distance between two objects which can be distinguished.

$$R = 0.61 \frac{\lambda}{NA}$$

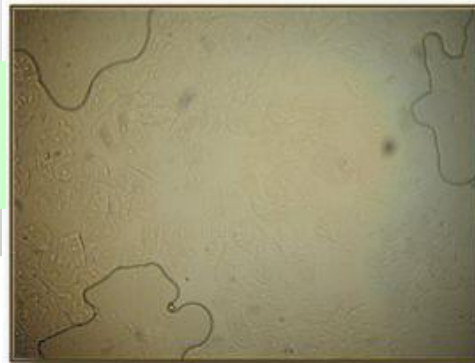
- NA = numerical aperture λ = wavelength

Observation Modes (B)



- **Phase Contrast (相差显微镜)**- is a type of light microscopy that enhances contrasts of transparent and colorless objects by influencing the optical path of light.
- The **Phase Contrast** microscope is able to show components in a cell or bacteria, which would be very difficult to see in an ordinary light microscope.

Normal
Microscope



Phase
Contrast

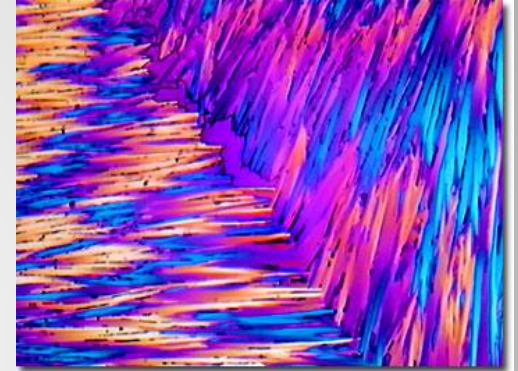


Observation Modes (C)

- **Polarization(偏正光)** - uses two polarizers, one on either side of the specimen, positioned perpendicular to each other so that only light that passes through the specimen reaches the eyepiece.

布洛芬

Ibuprofen



雌二醇

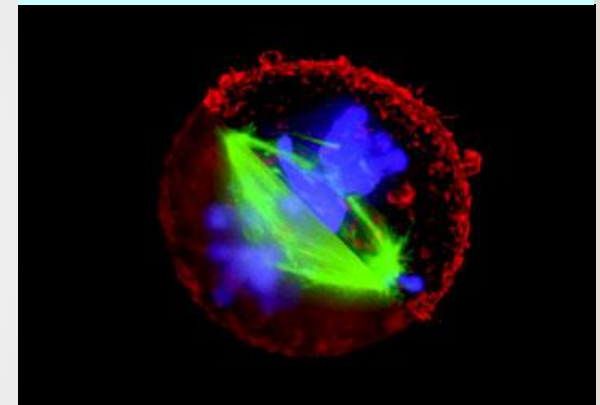
Estradiol



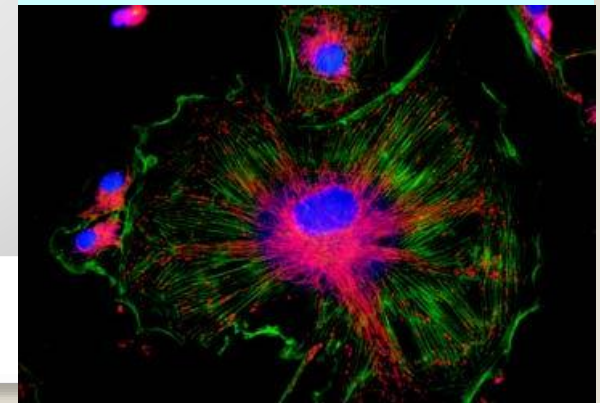
Observation Modes (D)

- **Fluorescence**- this technique is used to study specimens, which can be made to fluoresce.

Mitosis in hamster



Endothelium Cells



- Confocal Microscopy
共聚焦显微

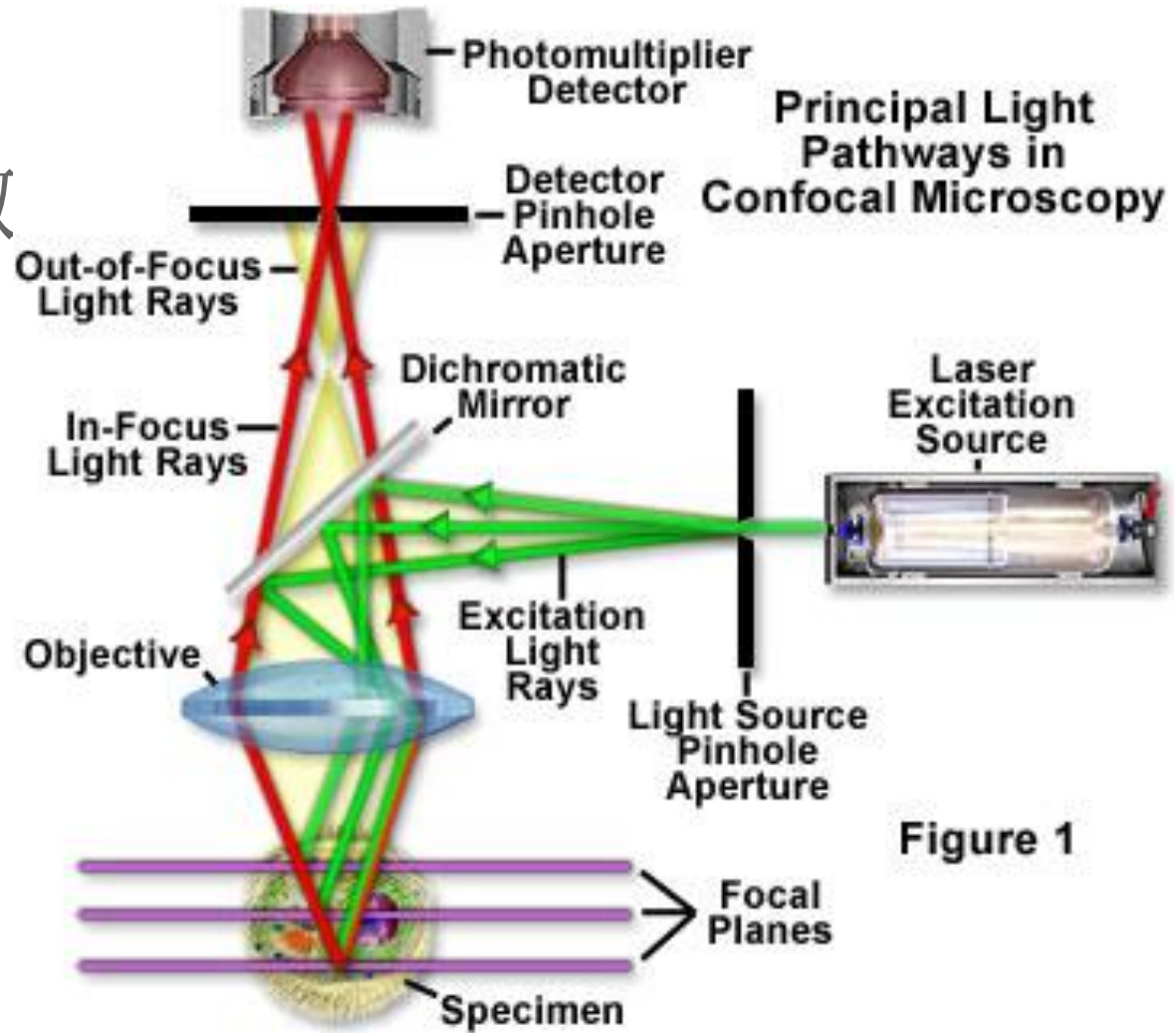
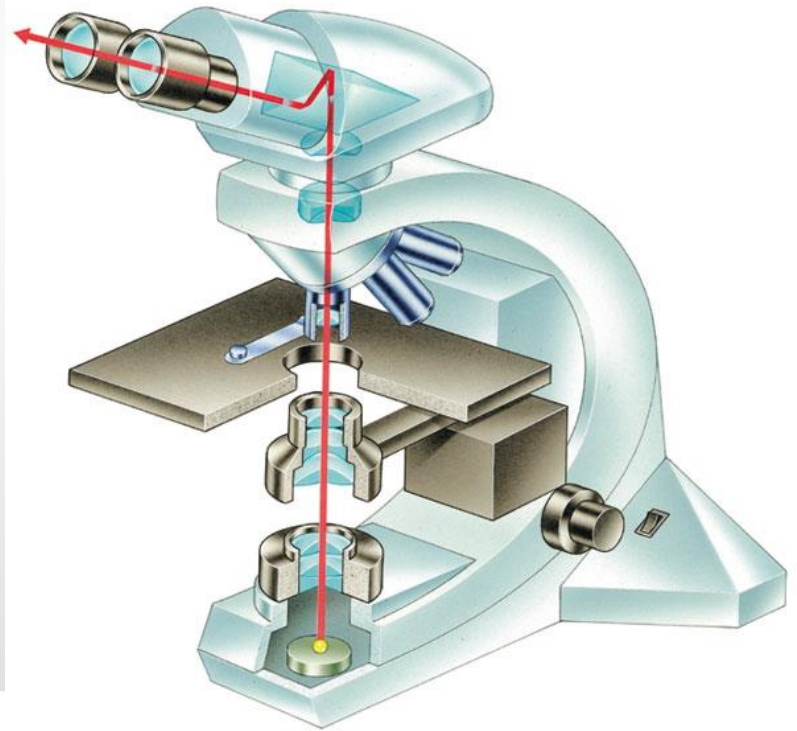


Figure 1

Advantages of Using Optical Microscopes

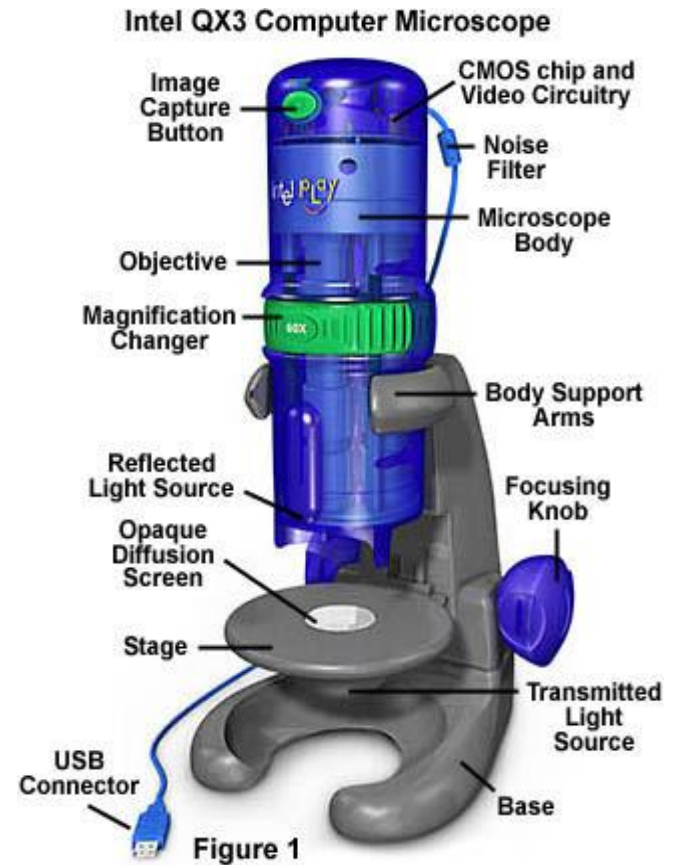
- **Inexpensive**
- **Easy to learn and operate**
- **Magnification acceptable up to 2,000 times**
- **Very sharp plane of focus**
- **Small and portable**



Expensive



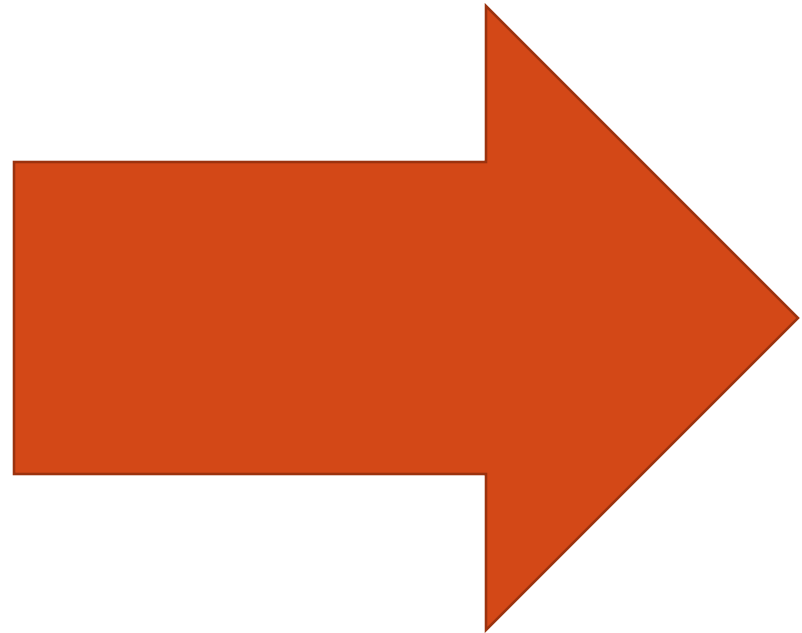
Cheap**



** If you can find one.

What can you do with it??

- Standard Imaging
 - Get some high detail images of things of interest – like bugs, microbes (the bigger ones) and pond scum creatures.





This is on your
disk

AMAZING MICROLIFE

by Shari Waters for

The Orlando Science Center

**Thank
You!**